Folha 6

Exercício 6.1

1

```
Integrate[3 x^2 - 2 x^5, x]
```

$x^3 - \frac{x^6}{3}$

2

```
Integrate[(Sqrt[x] + 2)^2, x]
```

$\frac{1}{6} x \left(24 + 16 \sqrt{x} + 3 x\right)$

3

```
Integrate[(2 x + 10)^20, x]
```

$\frac{1}{42} (10 + 2 x)^{21}$

4

```
Integrate[x^2 Exp[x^3], x]
```

$\frac{e^{x^3}}{3}$

5

```
Integrate[x^4 (x^5 + 10)^9, x]
```

$\frac{1}{50} \left(10 + x^5\right)^{10}$

6

```
Integrate[(2 x + 1) / (x^2 + x + 3), x]
```

$\text{Log}\left[3 + x + x^2\right]$

7

```
Integrate[Sqrt[2 x + 1], x]
```

$\frac{1}{3} (1 + 2 x)^{3/2}$

8

```
Integrate[x / (3 - x^2), x]
```

$-\frac{1}{2} \text{Log}\left[3 - x^2\right]$

9

```
Integrate[1 / (4 - 3 x), x]
```

$-\frac{1}{3} \text{Log}[4 - 3 x]$

10

```
Integrate[Exp[-3 x], x]
```

$-\frac{1}{3} e^{-3 x}$

11

```
Integrate[-7 / Sqrt[1 - 5 x], x]
```

$\frac{14}{5} \sqrt{1 - 5 x}$

12

```
Integrate[Sqrt[1 + 3 Log[x]] / x, x]
```

$\frac{2}{9} (1 + 3 \text{Log}[x])^{3/2}$

13

```
Integrate[x Sin[x^2], x]
```

$-\frac{1}{2} \text{Cos}[x^2]$

14

```
Integrate[1 / (x (Log[x]^2 + 1)), x]
```

ArcTan[Log[x]]

15

```
Integrate[(2 / x - 3)^2 / x^2, x]
```

$-\frac{4}{3 x^3} + \frac{6}{x^2} - \frac{9}{x}$

16

```
Integrate[Sin[Pi - 2 x], x]
```

$-\frac{1}{2} \text{Cos}[2 x]$

17

```
Integrate[Tanh[x], x]
```

Log[Cosh[x]]

18

```
Integrate[Sin[x] Cos[x], x]
```

$-\frac{1}{2} \text{Cos}[x]^2$

19

```
Integrate[Sin[2 x] Cos[x], x]
```

$-\frac{\text{Cos}[x]}{2} - \frac{1}{6} \text{Cos}[3 x]$

20

```
Integrate[Sin[x]^2, x]
```

$\frac{x}{2} - \frac{1}{4} \text{Sin}[2 x]$

21

```
Integrate[Sin[x / 2]^2 Cos[x / 2]^2, x]
```

$\frac{1}{4} \left(\frac{x}{2} - \frac{1}{4} \text{Sin}[2 x]\right)$

22

```
Integrate[Cos[x]^3, x]
```

$\frac{3 \text{Sin}[x]}{4} + \frac{1}{12} \text{Sin}[3 x]$

23

```
Integrate[x / (x^2 - 1), x]
```

$\frac{1}{2} \text{Log}[-1 + x^2]$

24

```
Integrate[x / Sqrt[x^2 - 1], x]
```

$\sqrt{-1 + x^2}$

25

```
Integrate[Sin[Log[x]] / x, x]
```

$-\mathrm{Cos}[\mathrm{Log}[x]]$

26

```
Integrate[-3 / (x Log[x] ^3), x]
```

$\frac{3}{2\,\mathrm{Log}[x]^2}$

27

```
Integrate[Exp[x] / (1 + Exp[2 x]), x]
```

$\mathrm{ArcTan}\left[e^x\right]$

28

```
Integrate[Exp[x] / (1 - 2 Exp[x]), x]
```

$-\frac{1}{2}\,\mathrm{Log}\left[1-2\,e^x\right]$

29

```
Integrate[Cos[7 x] ^-2, x]
```

$\frac{1}{7}\,\mathrm{Tan}[7\,x]$

30

```
Integrate[Sqrt[2 x - 1] - Sqrt[1 + 3 x], x]
```

$\frac{1}{3}\,(-1+2\,x)^{3/2}-\frac{2}{9}\,(1+3\,x)^{3/2}$

31

```
Integrate[(1 + Log[x] ^2) / x, x]
```

$\mathrm{Log}[x]+\frac{\mathrm{Log}[x]^3}{3}$

32

```
Integrate[(2 + Sqrt[ArcTan[2 x]]) / (1 + 4 x^2), x]
```

$\mathrm{ArcTan}[2\,x]+\frac{1}{3}\,\mathrm{ArcTan}[2\,x]^{3/2}$

33

```
Integrate[Exp[ArcTan[x]] / (1 + x^2), x]
```

$e^{\mathrm{ArcTan}[x]}$

34

```
Integrate[Sin[x] / Sqrt[1 + Cos[x]], x] // FullSimplify
```

$-2\,\sqrt{1+\mathrm{Cos}[x]}$

Exercício 6.2

a

```
Integrate[Log[x], x]
```

$-x+x\,\mathrm{Log}[x]$

b

```
Integrate[x Sin[2 x], x]
```

$-\frac{1}{2}\,x\,\mathrm{Cos}[2\,x]+\frac{1}{4}\,\mathrm{Sin}[2\,x]$

c

```
Integrate[ArcTan[x], x]
```

$x \text{ArcTan}[x] - \frac{1}{2} \text{Log}\left[1 + x^2\right]$

d

```
Integrate[x Cos[x], x]
```

$\text{Cos}[x] + x \text{Sin}[x]$

e

```
Integrate[Log[1 - x], x]
```

$-x - \text{Log}[1 - x] + x \text{Log}[1 - x]$

f

```
Integrate[x Log[x], x]
```

$-\frac{x^2}{4} + \frac{1}{2} x^2 \text{Log}[x]$

g

```
Integrate[x^2 Sin[x], x]
```

$-\left(-2 + x^2\right) \text{Cos}[x] + 2 x \text{Sin}[x]$

h

```
Integrate[x Sin[x] Cos[x], x]
```

$-\frac{1}{4} x \text{Cos}[2 x] + \frac{1}{8} \text{Sin}[2 x]$

i

```
Integrate[Log[x]^2, x]
```

$2 x - 2 x \text{Log}[x] + x \text{Log}[x]^2$

j

```
Integrate[Exp[x] Cos[x], x]
```

$\frac{1}{2} e^x (\text{Cos}[x] + \text{Sin}[x])$

k

```
Integrate[ArcSin[x], x]
```

$\sqrt{1 - x^2} + x \text{ArcSin}[x]$

l

```
Integrate[Exp[Sin[x]] Sin[x] Cos[x], x]
```

$e^{\text{Sin}[x]} (-1 + \text{Sin}[x])$

m

```
Integrate[ArcSin[Sqrt[x]] / Sqrt[x], x]
```

$2 \sqrt{1 - x} + 2 \sqrt{x} \text{ArcSin}\left[\sqrt{x}\right]$

n

```
Integrate[x ArcTan[x], x]
```

$-\frac{x}{2} + \frac{\text{ArcTan}[x]}{2} + \frac{1}{2} x^2 \text{ArcTan}[x]$

o

```
Integrate[x ArcTan[x], x]
```

$-\frac{x}{2} + \frac{\text{ArcTan}[x]}{2} + \frac{1}{2} x^2 \text{ArcTan}[x]$

p

```
Integrate[Sin[Log[x]], x]
```

$-\frac{1}{2} x \operatorname{Cos}[\operatorname{Log}[x]] + \frac{1}{2} x \operatorname{Sin}[\operatorname{Log}[x]]$

q

```
Integrate[Cosh[x] Sin[3 x], x]
```

$-\frac{3}{10} \operatorname{Cos}[3 x] \operatorname{Cosh}[x] + \frac{1}{10} \operatorname{Sin}[3 x] \operatorname{Sinh}[x]$

r

```
Integrate[x^3 Exp[x^2], x]
```

$\frac{1}{2} e^{x^2} \left(-1 + x^2\right)$

Exercício 6.3

a

```
Integrate[x (x + 3) ^ (1 / 3), x]
```

$\frac{3}{28} (3 + x)^{4/3} (-9 + 4 x)$

b

```
Integrate[1 / Sin[x], x]
```

$-\operatorname{Log}\left[\operatorname{Cos}\left[\frac{x}{2}\right]\right] + \operatorname{Log}\left[\operatorname{Sin}\left[\frac{x}{2}\right]\right]$

c

```
Integrate[x / Sqrt[2 - 3 x], x]
```

$-\frac{2}{27} \sqrt{2 - 3 x} (4 + 3 x)$

d

```
Integrate[Sin[Sqrt[x]] / Sqrt[x], x]
```

$-2 \operatorname{Cos}\left[\sqrt{x}\right]$

e

```
Integrate[Exp[2 x] / (3 + Exp[x]), x]
```

$e^{x} - 3 \operatorname{Log}\left[3 + e^{x}\right]$

f

```
Integrate[x^2 / Sqrt[1 - x^2], x]
```

$-\frac{1}{2} x \sqrt{1 - x^2} + \frac{\operatorname{ArcSin}[x]}{2}$

g

```
Integrate[Sqrt[x] / (x - x^(1 / 3)), x]
```

$2 \sqrt{x} + 3 \operatorname{ArcTan}\left[x^{1/6}\right] + \frac{3}{2} \operatorname{Log}\left[1 - x^{1/6}\right] - \frac{3}{2} \operatorname{Log}\left[1 + x^{1/6}\right]$

h

```
Integrate[Sqrt[1 + x^2], x]
```

$\frac{1}{2} \left(x \sqrt{1 + x^2} + \operatorname{ArcSinh}[x]\right)$

Exercício 6.4

a

```
Apart[(2 x^2 + x + 1) / ((x - 1) (x + 1) ^ 2)]
```

$\frac{1}{-1 + x} - \frac{1}{(1 + x)^2} + \frac{1}{1 + x}$

```
Integrate[(2 x^2 + x + 1) / ((x - 1) (x + 1) ^ 2), x]
```

$\frac{1}{1 + x} + \operatorname{Log}[-1 + x] + \operatorname{Log}[1 + x]$

b

```
Apart[(3 x^2 - 4 x - 1) / ((x^2 - 1) (x - 2))]
```

$$\frac{1}{-2+x}+\frac{1}{-1+x}+\frac{1}{1+x}$$

```
Integrate[(3 x^2 - 4 x - 1) / ((x^2 - 1) (x - 2)), x]
```

$$\mathrm{Log}\left[3+4\,(-2+x)+(-2+x)^2\right]+\mathrm{Log}[-2+x]$$

c

```
Apart[(2 x^2 - x - 2) / (x^2 (x - 2))]
```

$$\frac{1}{-2+x}+\frac{1}{x^2}+\frac{1}{x}$$

```
Integrate[(2 x^2 - x - 2) / (x^2 (x - 2)), x]
```

$$-\frac{1}{x}+\mathrm{Log}[2-x]+\mathrm{Log}[x]$$

d

```
Apart[(2 x^3 + 5 x^2 + 6 x + 2) / ((x (x + 1) ^3))]
```

$$\frac{2}{x}+\frac{1}{(1+x)^3}-\frac{1}{(1+x)^2}$$

```
Integrate[(2 x^3 + 5 x^2 + 6 x + 2) / ((x (x + 1) ^3)), x]
```

$$-\frac{1}{2\,(1+x)^2}+\frac{1}{1+x}+2\,\mathrm{Log}[x]$$

e

```
Apart[(x^2 - x + 2) / (x (x^2 - 1))]
```

$$\frac{1}{-1+x}-\frac{2}{x}+\frac{2}{1+x}$$

```
Integrate[(x^2 - x + 2) / (x (x^2 - 1)), x]
```

$$-2\,\mathrm{Log}[x]+\mathrm{Log}[1+x]+\mathrm{Log}\left[1-x^2\right]$$

f

```
Apart[(4 x^2 + x + 1) / (x^3 - x)]
```

$$\frac{3}{-1+x}-\frac{1}{x}+\frac{2}{1+x}$$

```
Integrate[(4 x^2 + x + 1) / (x^3 - x), x]
```

$$3\,\mathrm{Log}[1-x]-\mathrm{Log}[x]+2\,\mathrm{Log}[1+x]$$

g

```
Apart[27 / (x^4 - 3 x^3)]
```

$$\frac{1}{-3+x}-\frac{9}{x^3}-\frac{3}{x^2}-\frac{1}{x}$$

```
Integrate[27 / (x^4 - 3 x^3), x]
```

$$27\left(\frac{1}{6\,x^2}+\frac{1}{9\,x}+\frac{1}{27}\,\mathrm{Log}[3-x]-\frac{\mathrm{Log}[x]}{27}\right)$$

h

```
Apart[(x^4 - 8) / (x^3 - 2 x^2)]
```

$$2+\frac{2}{-2+x}+\frac{4}{x^2}+\frac{2}{x}+x$$

```
Integrate[(x^4 - 8) / (x^3 - 2 x^2), x]
```

$$-\frac{4}{x}+2\,x+\frac{x^2}{2}+2\,\mathrm{Log}[2-x]+2\,\mathrm{Log}[x]$$

i

```
Apart[(x + 3) / ((x - 2) (x^2 - 2 x + 5))]
```

$$\frac{1}{-2+x}+\frac{1-x}{5-2\,x+x^2}$$

```
Integrate[(x + 3) / ((x - 2) (x^2 - 2 x + 5)), x]
```

$-\frac{1}{2}\,\text{Log}\left[5+2\,(-2+x)+(-2+x)^2\right]+\text{Log}[-2+x]$

j

```
Apart[(x + 1) / (x (x^2 + 1) ^2)]
```

$\frac{1}{x}+\frac{1-x}{\left(1+x^2\right)^2}-\frac{x}{1+x^2}$

```
Integrate[(x + 1) / (x (x^2 + 1) ^2), x]
```

$\frac{1}{2}\left(\frac{1+x}{1+x^2}+\text{ArcTan}[x]+2\,\text{Log}[x]-\text{Log}\left[1+x^2\right]\right)$

k

```
Apart[(x + 2) / (2 x (x - 1) ^2 (x^2 + 1))]
```

$\frac{3}{4\,(-1+x)^2}-\frac{5}{4\,(-1+x)}+\frac{1}{x}+\frac{2+x}{4\left(1+x^2\right)}$

```
Integrate[(x + 2) / (2 x (x - 1) ^2 (x^2 + 1)), x]
```

$\frac{1}{2}\left(-\frac{3}{2\,(-1+x)}+\text{ArcTan}[x]-\frac{5}{2}\,\text{Log}[1-x]+2\,\text{Log}[x]+\frac{1}{4}\,\text{Log}\left[1+x^2\right]\right)$

l

```
Apart[(3 x^3 + x^2 - x - 1) / (x^2 (x^2 - 1))]
```

$\frac{1}{-1+x}+\frac{1}{x^2}+\frac{1}{x}+\frac{1}{1+x}$

```
Integrate[(3 x^3 + x^2 - x - 1) / (x^2 (x^2 - 1)), x]
```

$-\frac{1}{x}+\text{Log}[x]+\text{Log}\left[1-x^2\right]$

Exercício 6.5

a

```
Integrate[1 / ((2 + Sqrt[x]) ^7 Sqrt[x]), x]
```

$-\frac{1}{3\left(2+\sqrt{x}\right)^6}$

b

```
Integrate[Tan[x] ^2, x]
```

$-x+\text{Tan}[x]$

c

```
Integrate[(x + ArcSin[3 x] ^2) / Sqrt[1 - 9 x^2], x]
```

$-\frac{1}{9}\sqrt{1-9\,x^2}+\frac{1}{9}\,\text{ArcSin}[3\,x]^3$

d

```
Integrate[x Exp[Sqrt[1 - x^2]] / Sqrt[1 - x^2], x]
```

$-e^{\sqrt{1-x^2}}$

e

```
Integrate[Cos[x] ^-2 Sin[x] ^-2, x]
```

$-2\,\text{Cot}[2\,x]$

f

```
Integrate[Cos[x] ^2 Sin[x] ^2, x]
```

$\frac{x}{8}-\frac{1}{32}\,\text{Sin}[4\,x]$

g

```
Integrate[1 / (1 + Exp[x]), x]
```

$x-\text{Log}\left[1+e^x\right]$

```
Apart[1 / (t (1 + t)), t]
```

$\frac{1}{t} - \frac{1}{1+t}$

```
Integrate[1/t - 1/(1 + t), t]
```

Log[t] - Log[1 + t]

h

```
Integrate[1 / (x^2 Sqrt[4 - x^2]), x]
```

$-\frac{\sqrt{4-x^2}}{4x}$

Exercício 6.6

$-\left(-2+x^2\right)$ Cos[x] + 2 x Sin[x]

Exercício 6.7

a

$\frac{65}{6} - 8x - \frac{x^2}{2} + \frac{2x^3}{3}$

b

$\frac{5x}{4} - \frac{1}{8}$ Sin[2 x]

Exercício 6.8

a

```
Integrate[Exp[Pi x], {x, 0, 1}]
```

$\frac{-1+e^{\pi}}{\pi}$

b

```
Integrate[Abs[Sin[x]], {x, -Pi / 2, Pi / 2}]
```

2

c

```
Integrate[Abs[x - 1], {x, -3, 5}]
```

16

d

```
Integrate[Abs[(x - 1) (3 x - 2)], {x, 0, 2}]
```

$\frac{55}{27}$

e

```
Integrate[Sqrt[9 - x^2], {x, 0, 3}]
```

$\frac{9\pi}{4}$

f

```
Integrate[2 x Sqrt[4 - x], {x, -5, 0}]
```

$-\frac{1012}{15}$

g

```
Integrate[1 / (x^2 Sqrt[x^2 + 1]), {x, 3 / 4, 4 / 3}]
```

$\frac{5}{12}$

h

```
Integrate[Log[10, x^2 + 1], {x, 0, 1}]
```

$\frac{-4 + \pi + \text{Log}[4]}{\text{Log}[100]}$

i

```
Integrate[x^3 Exp[x^2], {x, 0, 2}]
```

$\frac{1}{2}\left(1 + 3\, e^4\right)$

j

```
Integrate[x Sin[x], {x, 0, Pi}]
```

π

k

```
Integrate[ArcSin[x], {x, 0, Sqrt[2] / 2}]
```

$-1 + \frac{4 + \pi}{4\sqrt{2}}$

l

```
Integrate[Sqrt[Abs[x]], {x, -3, 2}]
```

$\frac{2}{3}\left(2\sqrt{2} + 3\sqrt{3}\right)$

m

```
Integrate[Piecewise[{{x^2, 0 ≤ x ≤ 1}, {2 - x, 1 ≤ x ≤ 2}}], {x, 0, 2}]
```

$\frac{5}{6}$

n

```
Integrate[Piecewise[{{x, 0 ≤ x ≤ 1 / 2}, {-x, 1 / 2 ≤ x ≤ 1}}], {x, 0, 1}]
```

$-\frac{1}{4}$

Exercício 6.11

a

$$F'(x) = \frac{1}{\left(1 + x^2\right)^3}$$

b

$$F'(x) = \frac{2\,x}{\left(1 + x^4\right)^3}$$

c

$$F'(x) = \frac{2\,x^{13}}{1 + x^8} - \frac{3\,x^{20}}{1 + x^{12}}$$

Exercício 6.12

a

$$f(x) = 2\,x + 3\,x^2$$

b

$$f(x) = \frac{3\sqrt{x}\,e^{\sqrt{x}} + x\,e^{\sqrt{x}} - 4\,x}{2}$$

Exercício 6.13

a

-1

b

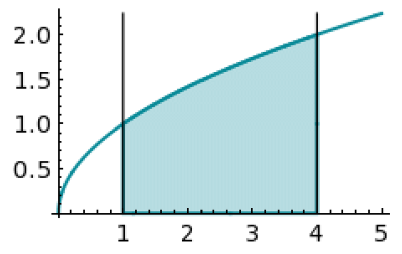
$-2\sqrt{3}$

Exercício 6.14

- **a** existe
- **b** não existe
- **c** não existe
- **d** existe
- **e** existe
- **f** existe

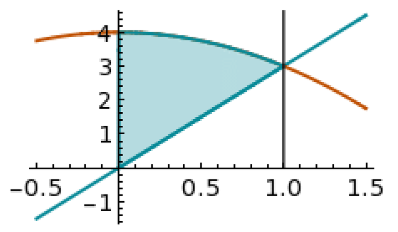
Exercício 6.15

a



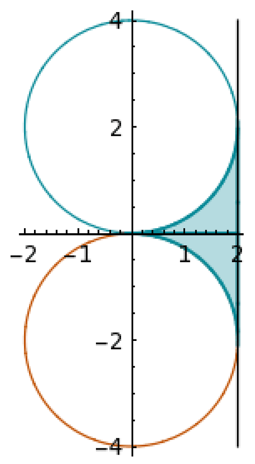
$\frac{14}{3}$

b



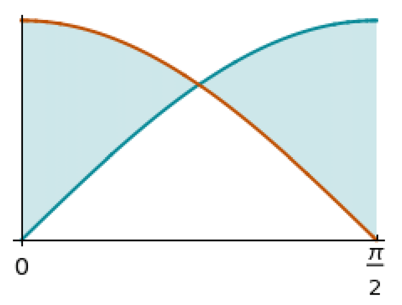
$\frac{13}{6}$

c



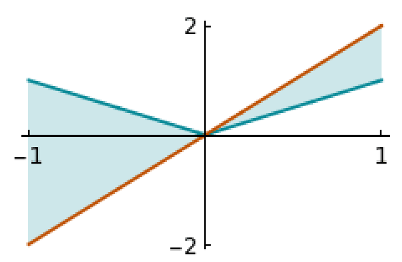
$8 - 2\pi$

d



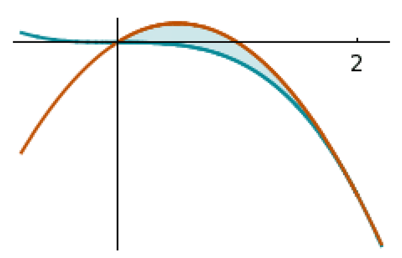
$2\left(-1+\sqrt{2}\right)$

e



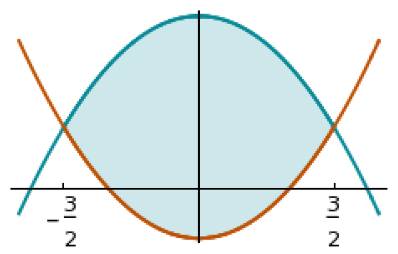
2

f



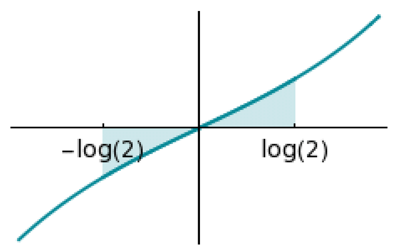
$\frac{4}{3}$

g



9

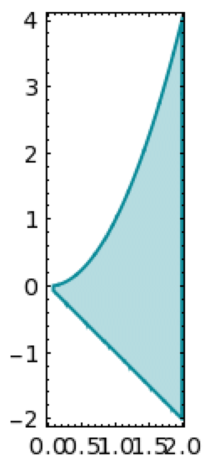
h



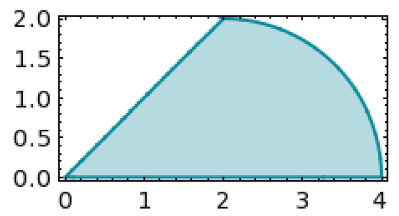
$\frac{1}{2}$

Exercício 6.16

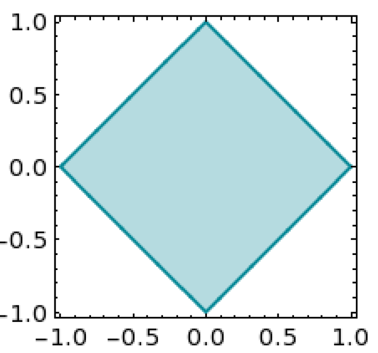
a



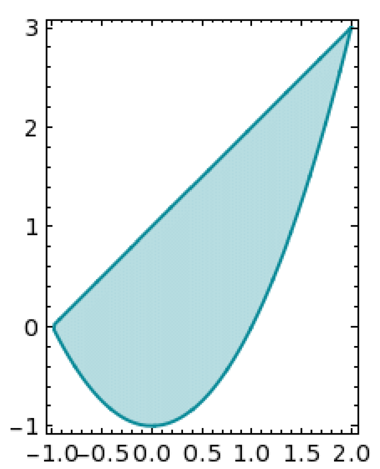
b



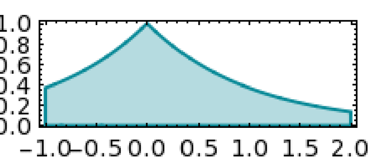
c



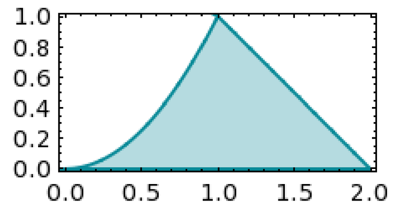
d



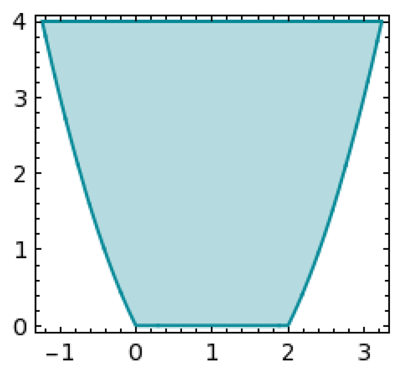
e



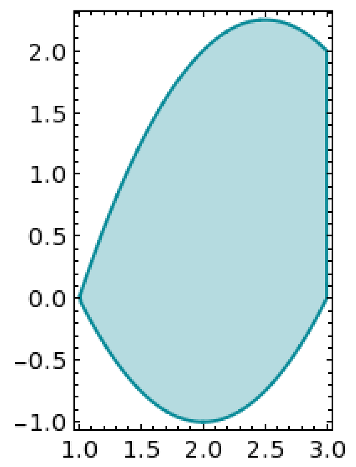
f


1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.5
1.0
1.5
2.0

g


4
3
2
1
0
−1
0
1
2
3

h


2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
1.0
1.5
2.0
2.5
3.0